FIITJEE

ALL INDIA TEST SERIES

FULL TEST – IX

JEE (Advanced)-2020

PAPER – 1

TEST DATE: 16-09-2020

Time Allotted: 3 Hours **Maximum Marks: 186**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-C.**

1. ***Section–A (01 – 04, 19 – 22, 37 – 40)*** *contains 12 multiple choice questions which out of 4 options have* ***only one correct answer****. Each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

   Section-A (05 – 12, 23 – 30, 41 – 48) *contains 24 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for all correct answer.*

   *Full Marks : **+4** If only (all) the four option(s) is (are) chosen.*
   *Partial Marks : **+3** If all the four options are correct but ONLY three options are chosen.*
   *Partial Marks : **+2** If three or more options are correct but ONLY two options are chosen and both of which are correct options.*
   *Partial Marks : **+1** If two or more options are correct but ONLY one option is chosen and it is a correct option.*
   *Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).*
   *Negative Marks : **–1** In all other cases.*

2. ***Section-C (13 – 18, 31 – 36, 49 – 54)*** *contains 18 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Physics

PART – I

SECTION – A
(One Options Correct Type)

This section contains **04 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A ray of light moving along vector $\left(5\sqrt{2}\hat{i}-5\hat{j}-5\hat{k}\right)$ undergoes refraction at an interface of two media, which is y-z plane. The refractive index for $x \le 0$ is $\sqrt{3}$ while for $x \ge 0$ it is $\sqrt{2}$. Then the unit vector along the refracted ray is

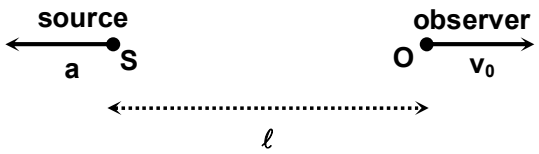
(A) $\dfrac{\sqrt{3}\hat{i}-\sqrt{2}\hat{j}-\sqrt{2}\hat{k}}{\sqrt{8}}$

(B) $\dfrac{\sqrt{3}\hat{i}-\sqrt{3}\hat{j}-\sqrt{2}\hat{k}}{\sqrt{8}}$

(C) $\dfrac{\sqrt{2}\hat{i}-\sqrt{3}\hat{j}-\sqrt{3}\hat{k}}{\sqrt{8}}$

(D) $\dfrac{\sqrt{2}\hat{i}+\sqrt{3}\hat{j}-\sqrt{3}\hat{k}}{\sqrt{8}}$

2. At t = 0, source starts accelerating with an acceleration a and observer starts moving with constant velocity v_0 as shown in the figure simultaneously. Source emits a frequency f and velocity of sound in the air is v. The frequency of the initial sound as detected by the observer is



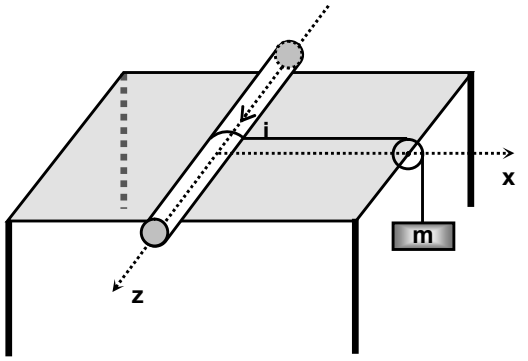
(A) $\dfrac{(v-v_0)f^2}{(2vf+a)}$

(B) $\dfrac{2(v-v_0)f^2}{(2vf+a)}$

(C) $\dfrac{(v-v_0)f^2}{(2vf-a)}$

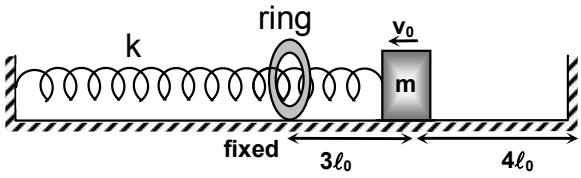
(D) $\dfrac{2(v-v_0)f^2}{(2vf-a)}$

3. A conducting rod of length 'ℓ' = $2\sqrt{5}$ meter and mass 'm' = 4 kg lies on the horizontal table and connected with a hanging block of mass 4 kg by an ideal string passing over the pulley as shown in the figure. Coefficient of friction between the rod and the table is 'μ' $=\left(\dfrac{1}{2}\right)$. If the current in the conductor is 2 ampere, then find the minimum magnitude of magnetic field strength (in Tesla) such that conducting rod just starts to translate along negative x-axis. (take g = 10 m/sec^2) [neglect the radius of rod]



(A) 6
(B) 2
(C) 4
(D) 8

4. Spring is in its natural length. Collision of the block with the ring fixed with the horizontal surface and wall is perfectly elastic. Find the time period of oscillation of block if at t = 0, a velocity $v_0 = 5\ell_0\sqrt{\frac{k}{m}}$ is given to the block.



(A) $2\pi\sqrt{\frac{m}{k}}$

(B) $\frac{\pi}{2}\sqrt{\frac{m}{k}}$

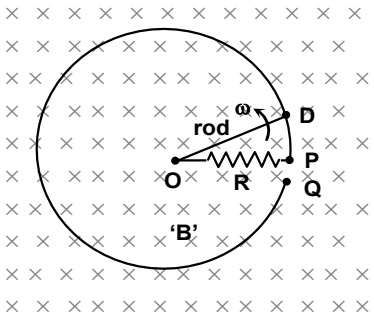
(C) $\pi\sqrt{\frac{m}{k}}$

(D) $\frac{3\pi}{2}\sqrt{\frac{m}{k}}$

(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

5. A conducting rod OD of mass m and length ℓ rotates about one end O, with its other end sliding smoothly on a circular conducting ring, as shown in the figure. A constant and uniform magnetic field 'B' acts into the plane of the ring and rod OD rotates with a constant angular velocity ω by applying some external force. Assume that the space is gravity free and that there is electrical contact between the ring and the rod at D. The ring and the rod have negligible resistance. There is a small gap in the ring between P and Q, so that the current flows along a unique path. Then,



(A) The torque due to external agent about O is $\frac{B^2\omega\ell^4}{2R}$

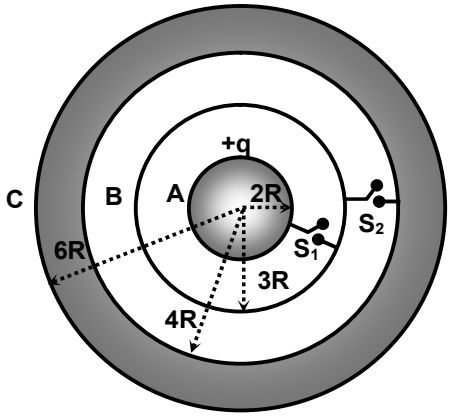
(B) The power delivered by the external agent is $\frac{B^2\omega^2\ell^4}{4R}$

(C) The rate of heat rejected by the resistor R is $\frac{B^2\omega^2\ell^4}{4R}$

(D) Work done by external agent for half revolution of rod is $\frac{\pi B^2\omega\ell^4}{4R}$

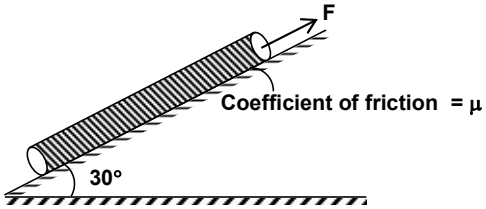
6. The magnitude of angular momentum, kinetic energy, velocity and frequency of an electron in a hydrogen atom corresponding to the quantum number n are L, K, V and f respectively. Then according to Bohr's theory of hydrogen atom:
   (A) $K^{-1}Lf$ is constant for the all orbits.
   (B) Kvf^{-1} is constant for the all orbits.
   (C) $K^{-1}Lv$ is proportional to n^2.
   (D) $K^{-1}f$ is proportional to 1/n.

7. Three conducting bodies A(solid sphere), B(thin spherical shell) and C(hollow sphere with thickness 2R) are arranged as shown in the figure. A charge q is given to the inner sphere. Choose the correct option(s).



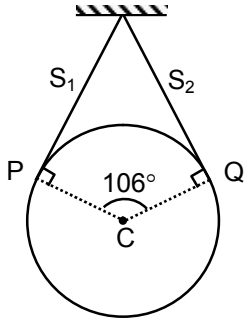
   (A) When S_1, S_2 both are open capacity of the system is C_1, then C_1 is $\frac{48\pi\varepsilon_0 R}{5}$
   (B) When S_1 is closed but S_2 is open capacity is C_2, then C_2 is $16\pi\varepsilon_0 R$
   (C) When S_2 is closed but S_1 is opened capacity is C_3, then C_3 is $12\pi\varepsilon_0 R$
   (D) When both S_1 and S_2 are closed. Capacity is C_4, then C_4 is $24\pi\varepsilon_0 R$

8. A uniform wire of mass m, length L and cross-sectional area 'A', made of a material of Young's modulus Y, is pulled by a constant force F on an inclined plane such that it move upto the inclined plane as shown in figure. Choose the correct option(s).



   (A) The extension in the wire is $\frac{F\ell}{2YA}$.
   (B) The tension in the rod at a distance $\frac{L}{4}$ from the lower end of the rod is $\frac{F}{4}$.
   (C) The elongation potential energy (elastic) in the rod is $\frac{1}{6}\frac{F^2L}{YA}$.
   (D) The elongation potential energy (elastic) in the rod is $\frac{1}{4}\frac{F^2L}{YA}$.

9. A solid sphere of mass 'm' and radius 'R' is held in equilibrium by two identical strings 'S_1' and 'S_2' of negligible mass as shown in the figure. Choose the correct option(s).



(A) Tension in the string S_2 when solid sphere is in equilibrium is $\frac{5}{8}mg$.

(B) Acceleration of point P just after cutting the string S_2 is $\frac{3}{5}g$.

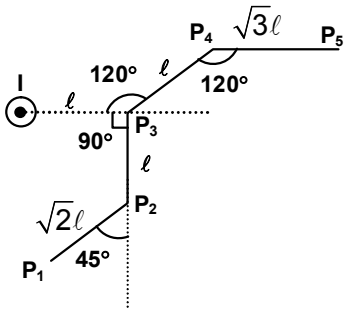
(C) Tension in the string 'S_1' immediately after cutting the string 'S_2' is $\frac{8mg}{35}$.

(D) Horizontal component of acceleration of centre of mass of solid sphere immediately after cutting the string 'S_2' is $\frac{24}{175}g$.

10. One mole of an ideal monoatomic gas undergoes a process $T = \left(\frac{T_0}{V_0^2}\right)V^2$, where T_0 and V_0 are the initial temperature and volume of the gas. If volume of gas changes from V_0 to $2V_0$. Then choose the correct option(s).

(A) The amount of heat transferred to the gas is $6RT_0$.
(B) The molar heat capacity of the gas is 2R.
(C) Final temperature of the gas is $4T_0$.
(D) The graph between the temperature and volume is parabolic.

11. A straight conducting wire of infinite length lies perpendicular to the plane of the paper and current of magnitude 'I' is flowing out to the paper as shown in figure. Then choose the correct option(s).



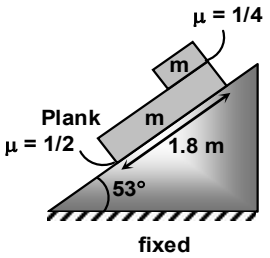
(A) The value of $\int_{P_1P_2} \vec{B}\cdot\overrightarrow{d\ell} = \frac{\mu_0 I}{8}$.

(B) The value of $\int_{P_1P_2} \vec{B}\cdot\overrightarrow{d\ell} = \int_{P_2P_3} \vec{B}\cdot\overrightarrow{d\ell}$.

(C) The value of $\int_{P_3P_4} \vec{B}\cdot\overrightarrow{d\ell} = 2\int_{P_4P_5} \vec{B}\cdot\overrightarrow{d\ell}$.

(D) The value of $\int_{P_1P_2P_3P_4P_5} \vec{B}\cdot\overrightarrow{d\ell} = \frac{7\mu_0 I}{24}$.

12. A plank of length 1.8 m and a small block of same mass initially at rest on a fixed large wedge as shown in the figure. The coefficient of friction between the plank and block is $\frac{1}{4}$ and between the plank and the wedge is $\frac{1}{2}$. Both block and plank are released from rest at t = 0. Then choose the correct option(s). (Take g = 10 m/s^2)



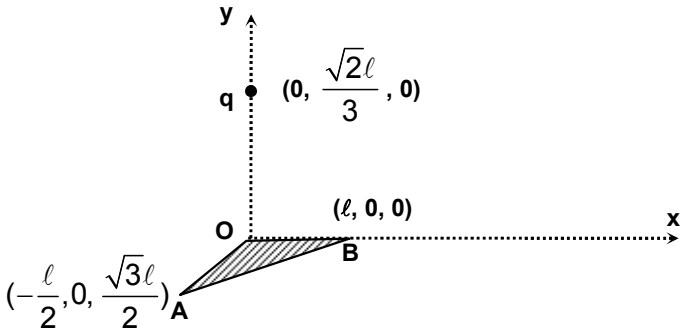
(A) Acceleration of the block with respect to ground is 6.5 m/s^2.
(B) Acceleration of the plank with respect to ground is 3.5 m/s^2.
(C) The time when the block just drops down from the plank is $\sqrt{1.2}$ sec.
(D) Displacement of the block with respect to the ground when the block just drops down from the plank is 3.9 m.

SECTION – C
(Numerical Answer Type)

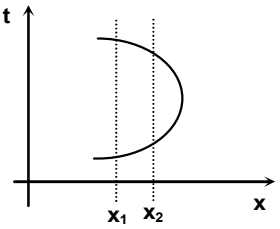
This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

13. The pitch of a screw gauge is 1 mm and its circular scale is divided into 100 divisions. When nothing is put between the studs the zero of main scale is not seen, but when circular scale is rotated by 270° the zero of main scale is seen and the zero of main scale coincide with the 12th division of circular scale. When a glass plate is placed between the studs, the circular scale lies between 24th to 25th division of main scale and circular scale reads 26 divisions. Find the thickness of the glass plate in millimeter. (Assume value of one main scale division is same as pitch)

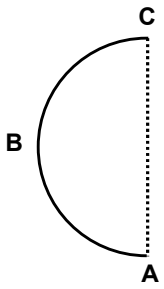
14. A point charge 'q' is placed at a distance $\frac{\sqrt{2}\ell}{3}$ (on the y-axis) vertically above the x-z plane. The flux of electric field passing through the triangular plate which lies in the x-z plane as shown in figure, is $\frac{q}{10m\varepsilon_0}$. Find the value of m.



15. The acceleration in a space is a_0 m/s^2 towards negative x-axis, where a_0 is a positive constant. A ball is projected and it passes through two given points x_1, x_2 in both the directions. If the time the ball takes to pass x_1 = 3.10m in both the directions is t_1 = 4 sec and time to pass x_2 = 10.75m in both the directions is t_2 = 2sec. Then find the magnitude of a_0.



16. A semicircular conducting ring ABC of mass m = 20 kg and radius R = 4m is placed on a horizontal surface and a uniform magnetic field B = 2 Tesla exists parallel to horizontal surface but perpendicular to the line AC. Almost instantaneously, a certain amount of charge is passed through it, then it is found to jump to a height h = 5m. Then, find the amount of charge passed through the conductor in coulomb. (take g = 10 m/s^2)



17. Suppose that one of the slits in YDSE is wider than the other, so the amplitude of the light reaching on screen from one slit, acting alone, is thrice that from the other slit, acting alone. The intensity for Central Bright Fringe is I_0 = 5 W/m^2. Then find the intensity I (W/m^2) on the screen for (angular position) $\theta = 37°$. (Take $d/\lambda = 5/6$, where d is the separation between the slits and λ is the wavelength of the light)

18. A certain radio nuclide starts getting produced at a constant rate 10^4 sec^{-1} by some mechanism at a certain moment t = 0. At this moment there are already 10^5 number of radio nuclides present in the sample. The mean life of this radio-nuclide is 20 seconds. At t = 40 sec, the $\frac{dN}{dt}$ is $\frac{10^4}{K}$ (in sec^{-1}). Then find the value of K. (Take $e^2 = 7.4$)

Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **04 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

19. The correct order of thermal stability of the following hydrides is:
    (A) $H_2O < H_2S > H_2Se > H_2Te > H_2Po$
    (B) $NH_3 < PH_3 < AsH_3 < SbH_3 > BiH_3$
    (C) $H_2O > H_2S > H_2Se > H_2Te > H_2Po$
    (D) $PH_3 < AsH_3 < SbH_3 < BiH_3 < NH_3$

20. Elevation in boiling point of an aqueous solution of a non-electrolyte solute is 1.56^o, what is the depression in freezing point of this solution.
    $K_f(H_2O) = 1.86^o\ mol^{-1}\ kg$
    $K_b(H_2O) = 0.52^o\ mol^{-1}\ kg$
    (A) 3.12
    (B) 3.58
    (C) 5.58
    (D) 3.46

21. Average velocity at T_1 of CO is equal to most probable velocity at T_2. Thus, T_1/T_2 is
    (A) 0.785
    (B) 1.18
    (C) 0.079
    (D) 2.36

22. In a vessel 0.2 mole of each He, N_2 and O_2 was present. Pinhole of vessel opened in vacuum for a short time period. What will be the order of partial pressure of gases inside the vessel now?
    (A) $P_{O_2} > P_{He} > P_{N_2}$
    (B) $P_{N_2} > P_{O_2} > P_{He}$
    (C) $P_{He} > P_{N_2} > P_{O_2}$
    (D) $P_{He} < P_{N_2} < P_{O_2}$

(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

23. Which of the following statement(s) is/are correct?
    (A) $\overline{KE}_{molecule}$ of a triatomic linear molecule is $\frac{5}{2}KT$
    (B) $\left(\frac{\partial G}{\partial T}\right)_P = -S$
    (C) In adiabatic free expression of an ideal gas, temperature of gas also not changed.
    (D) $\Delta S_{univ} = 0$ for irreversible adiabatic expansion.

24. Consider the following equilibrium?

$N_2(g) + 3H_2(g) \overset{200^\circ C}{\rightleftharpoons} 2NH_3(g)$ $\quad \Delta H = -92$ kg / mol

Correct statement(s) is/are:

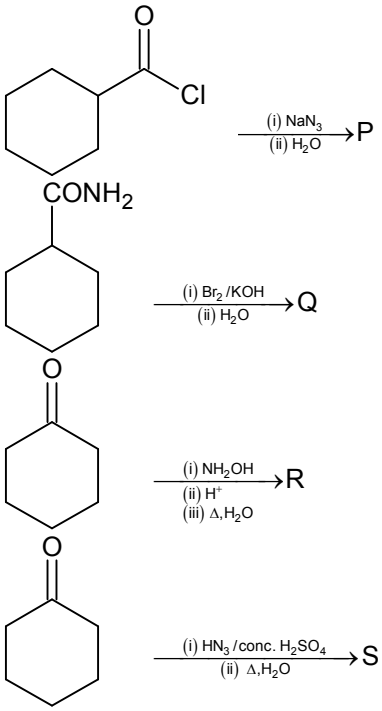
(A) On adding $N_2(g)$ in the equilibrium mixture. Equilibrium shift to forward direction due to increase in entropy in forward direction.

(B) At temp. < 200°C reaction becomes faster.

(C) On increasing the pressure yield of NH_3 increase.

(D) No effect of introduction of inert gas at equilibrium mixture.

25. Which of the following products are identical?



(A) P, Q, R, S

(B) P, Q

(C) R, S

(D) Q, R

26. Correct reaction with products are:

(i) $NH_4NO_2 + NaOH + Zn \text{ powder} / \Delta \longrightarrow NH_3 + Na_2ZnO_2$

(ii) $NH_4Cl + NaNO_3 \xrightarrow{\Delta} N_2O + NaCl + H_2O$

(iii) $(NH_4)_2 Cr_2O_7 \xrightarrow{\Delta} N_2 + 4H_2O + Cr_2O_3$

(iv) $NH_2CONH_2 \xrightarrow[HCl]{NaNO_2} NH_3 + CO_2 \uparrow + NaCl$

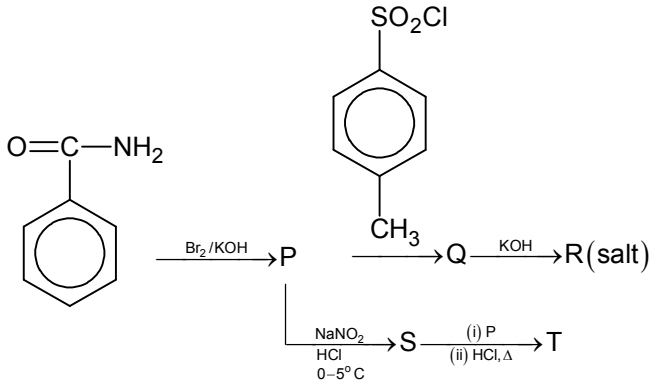
(A) ii, iii

(B) ii, iv

(C) i, iv

(D) i, iii

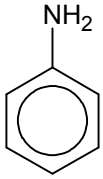
27.



Which of the following is(are) correct?

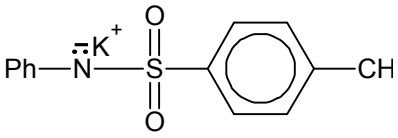
(A)

P =



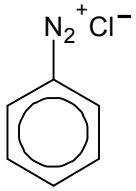
(B)

R is



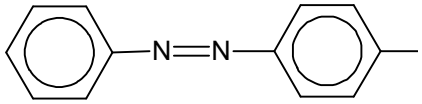
(C)

S is



(D)

T is



28. Which of the following give ppt. when H_2S is passed through the ammonical solution of their salts (in presence of NH_4Cl)
(A) Cu^{2+}
(B) Bi^{3+}
(C) Zn^{2+}
(D) Mg^{2+}

29. Which of the following ore(s) is/are concentrated by froth floatation method?
(A) Galena
(B) Chalcocite
(C) Bauxite
(D) Cinnabar

30. $\text{A salt's solution} \xrightarrow[\text{Excess}]{\text{NaOH}} \underset{\text{ppt.}}{\text{P}}\downarrow + \text{Q}_{\text{solution}}$

$\text{Q}_{\text{solution}} \xrightarrow[\text{H}_2\text{SO}_4]{\text{K}_2\text{Cr}_2\text{O}_7} \underset{\text{Red gas}}{\text{R}}\uparrow \xrightarrow{\text{NaOH}} \underset{\text{Yellow soln.}}{\text{S}} \xrightarrow{\text{AgNO}_3} \underset{\text{Brick red}}{\text{T}}\downarrow$

$\text{P} \xrightarrow[\text{ii) KI}]{\text{i) conc. HCl}} \underset{\text{Black}}{\text{U}}\downarrow \xrightarrow[\text{Excess}]{\text{KI}} \underset{\text{Soluble complex orange}}{\text{V}}$

Which of the following is/are correct according to the above reaction?

(A) R is CrO_2Cl_2

(B) T is Ag_2CrO_4

(C) U is BiI_3

(D) P is $Bi(OH)_3$

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

31. Henry's law constants for O_2 and N_2 dissolved in water at 273 K are 2.03×10^8 Pa and 5.07×10^8 Pa. respectively. A sample of water at a temperature just above 273 K was equilibrated with air (21% oxygen and 79% N_2) at 1 atm. The dissolved gas was separated from a sample of this water and then dried. Determine the composition of N_2 in percentage.

32. A solution containing 0.624 gm of naphthalene (molar mass 128.2 gm mol^{-1}) in 60 gm CCl_4. Yields a boiling point elevation of 0.42 K. While a solution of 0.53 gm of an unknown solute in the same mass of solvent gives a boiling point elevation of 0.75 K. Find the molar mass of the unknown solute.

33. Two containers A and B of identical volume. Container 'A' have 2 moles of N_2 and 3 mole of He. Container B has 5 mole of O_2. Both containers are at same temperature. Calculate the average rate of effusion of mixture of O_2.

34. Calculate the magnitude of work done (in kJ) in isothermal reversible expansion of 1 m^3 of He gas ($C_V = \frac{3}{2}R$) at 273.15 K and 10 atm to final pressure of 1 atm (in kJ)

35. For equilibrium

$\text{LiCl.3NH}_3\,(\text{s}) \rightleftharpoons \text{LiCl.NH}_3\,(\text{s}) + 2\text{NH}_3\,(\text{g}),\ K_P = 16\ \text{atm}^2$ at 50°C. A 5 litre vessel contains 0.2 mole of LiCl.NH_3(s). How many moles of NH_3 should be added to the flask at this temperature to drive the backward reaction to completion: [R = 0.0821 litre atm mol^{-1} K^{-1}]

36. The radius of an He atom is 1.23×10^{-9} cm. A 200 cm^3 container is filled with Xe at a pressure of 1 atm and a temperature of 273 K. The % of volume that is occupied by He atom is $x \times 10^{-7}$, value of x is:

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **04 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

37. Let A and B be any two points on y-axis and the line y = 2x respectively. Also P and Q be the feet of perpendicular from A and B on the line y = 2x and y-axis respectively. If AB passes through the points (1, 3), then the line PQ passes through fixed point (x_1, y_1) such that $\frac{y_1}{x_1}$ is

(A) 5
(B) 6
(C) 7
(D) 8

38. The value of the integral $\int_0^{\pi/2} \frac{\sin 40x}{\sin 5x} dx$ is

(A) $\frac{1}{5}\left(1-\frac{1}{3}+\frac{1}{5}-\frac{1}{7}\right)$

(B) $\frac{2}{5}\left(1-\frac{1}{3}+\frac{1}{5}-\frac{1}{7}\right)$

(C) $\frac{1}{5}\left(1+\frac{1}{3}+\frac{1}{5}+\frac{1}{7}\right)$

(D) $\frac{2}{5}\left(1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}\right)$

39. Let f(x) be a differentiable function satisfying $e^{2x} f(x) = -\frac{1}{2}\tan x + \int_0^x e^{2t} \tan(x-t) dt$, then

(A) f(x) is increasing $\forall\ x \in \left[0, \frac{\pi}{8}\right]$

(B) f(x) is decreasing $\forall\ x \in \left[\frac{\pi}{6}, \frac{\pi}{3}\right]$

(C) f(x) is increasing $\forall\ x \in \left[\frac{\pi}{4}, \frac{\pi}{3}\right]$

(D) f(x) is decreasing $\forall\ x \in \left[\frac{\pi}{3}, \frac{\pi}{2}\right)$

40. In ΔABC, $\angle A = \tan^{-1} 2$, $\angle C = \tan^{-1} 3$. Let D be an interior point on side AC such that area of ΔABD is 4.5 times that of area of ΔBCD. If $\angle ABD = \theta$, then the value of $\sin 2\theta$ is

(A) $\frac{7}{24}$

(B) $\frac{24}{25}$

(C) $\frac{7}{25}$

(D) $\frac{3}{5}$

(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

41. Let DEF be the triangle formed by joining the foot of altitudes from vertices A, B and C on sides BC, CA and AB respectively of ΔABC. If H be the orthocentre of ΔABC and P, Q be the foot of perpendicular from D on the altitudes BE and CF respectively, then

(A) $\frac{DP}{EH} + \frac{DQ}{FH} = \tan A$

(B) $\frac{DP}{EH} + \frac{DQ}{FH} = \cot A$

(C) $\frac{DP}{EH} + \frac{DQ}{FH} = \sqrt{2} + 1$, if $\angle A = \frac{3\pi}{8}$

(D) $\frac{DP}{EH} + \frac{DQ}{FH} = \sqrt{2} + 1$, if $\angle A = \frac{\pi}{8}$

42. Let $I_1 = \int_0^{\frac{\pi}{200}} \sin\left(2\pi\cos^2 100x\right)dx$, $I_2 = \int_0^{\frac{\pi}{200}} \sin\left(\pi\cos^2 100x\right)dx$, $I_3 = \int_0^{\frac{\pi}{200}} \sin\left(\pi\cos^2 50x\right)dx$, then

(A) $I_1 = 0$

(B) $I_2 = 0$

(C) $I_1 = I_2$

(D) $I_2 = I_3$

43. Let $f(x) = \sin^{-1}\left(\frac{4-x^2}{4+x^2}\right) + \cos^{-1}\left(\frac{4x}{4+x^2}\right)$, then

(A) total number of local extremum of f(x) in (–4, 4) are 1

(B) total number of local maximum of f(x) in (–4, 4) are zero

(C) maximum value of f(x) in (–4, 4) is π

(D) range of f(x) in (–4, 4) is $\left(-\frac{\pi}{2}, \pi\right]$

44. Let a ray of light through P(10, 25, 35) is incident a point Q on the plane x + y + z = 100 and passes through the point R(40, 60, 60) after reflection, then

(A) coordinates of foot of perpendicular from P on the given plane is (20, 45, 35)

(B) equation of incident ray is $\frac{x-10}{2}=\frac{y-25}{1}=\frac{z-35}{3}$

(C) equation of reflected ray is $\frac{x-40}{2}=\frac{y-60}{3}=\frac{z-60}{1}$

(D) equation of plane containing incident ray and reflected ray $2x - y - z + 40 = 0$

45. Let f: [0, 8] $\to$ R be a differentiable function such that f(0) = 0, f(4) = 2, f(8) = 4, then which of the following is/are correct?

(A) There exist some $C_1, C_2 \in (0, 8)$ such that $f'(C_1)f(C_2) = \frac{3}{2}$

(B) There exist some $C_1, C_2 \in (0, 8)$ such that $f'(C_1)f(C_2) = \frac{7}{2}$

(C) There exist some $\alpha, \beta \in (0, 4)$ such that $3\left(\sqrt{\alpha}f\left(\alpha\sqrt{\alpha}\right)+\sqrt{\beta}f\left(\beta\sqrt{\beta}\right)\right)=\int_0^8 f(t)\,dt$

(D) There exist some $\alpha, \beta \in (0, 4)$ such that $2\left(\alpha f\left(\alpha^2\right)+\beta f\left(\beta^2\right)\right)=\int_0^{16} f(t)\,dt$

46. Let α and $f(\alpha)$ be the eccentricity of the ellipse $\frac{x^2}{3b^2-2a^2}+\frac{y^2}{2b^2-a^2}=1$; $(b^2 > a^2)$ and $\frac{x^2}{2b^2-a^2}+\frac{y^2}{b^2}=1$; then

(A) $f(\alpha)=\frac{\alpha}{\sqrt{1-\alpha^2}}$

(B) $\int_0^{1/2} ffff(\alpha)\,d\alpha=\frac{1}{4}$

(C) $\int e^{\alpha}\left(f(\alpha)-f''(\alpha)\right)d\alpha=e^{\alpha}\left[\frac{1}{\left(1-\alpha^2\right)^{3/2}}+\frac{\alpha}{\sqrt{1-\alpha^2}}\right]+c$

(D) $\underbrace{ffff.....f(\alpha)}_{2018 \text{ times}}=\frac{\alpha}{\sqrt{1-2018\alpha^2}}$

47. Three alternate numbers a, b, c are removed from n numbers listed 1, 3, 5, 7, 9, ... If the arithmetic mean of the remaining numbers is $\frac{105}{2}$, then which of the following option(s) is/are correct?

(A) sum of the removed numbers is 81

(B) sum of the removed numbers is 61

(C) number of integral solutions of the equation xyz = n are 36

(D) number of positive integral solutions of the equation x + y + z = n : (where x, y, z are odd) is 325

48. Let A be a square matrix of order 3 which satisfies the equation $A^4 - 8A^3 + 24A^2 - 37A + 16I = 0$ and $B = A - 2I$ be another matrix. If Det A = 5 and Det B > 0, A^{-1} represent inverse of matrix A, then

(A) $\text{Det}(\text{Adj}\,(I - 2A^{-1}))$ is equal to 1

(B) $\text{Det}\left(\text{Adj}\left(\frac{B}{5}\right)^{-1}\right)$ is equal to 625

(C) $\text{Det}(I + 2B^{-1})$ is equal to 1

(D) Det B = 25

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

49. Let OABC be a regular tetrahedron with $\overrightarrow{OA} = \vec{a}$, $\overrightarrow{OB} = \vec{b}$, $\overrightarrow{OC} = \vec{c}$ and having volume $\frac{\sqrt{3}}{2\sqrt{2}}$. If $\hat{p}, \hat{q}$ and $\hat{r}$ are unit vectors along bisectors of angle between pair of edges represented by $\overrightarrow{OA}$, $\overrightarrow{OB}$; $\overrightarrow{OB}$, $\overrightarrow{OC}$ and $\overrightarrow{OC}$, $\overrightarrow{OA}$ respectively, then the value of $27\left[\hat{p}\times\hat{q} \quad \hat{q}\times\hat{r} \quad \hat{r}\times\hat{p}\right]$ is

50. Let abc be a three digit number whose digits are in G.P. On subtracting 100 from the number, the digits of the three digit number obtained form an A.P. The sum of all such 3 digit numbers possible is equal to

51. Let a, b be the roots of the equation $x^2 - 10x - 8 = 0$. If $S_n = a^n b + b^n a$; $n \in N$, then the value of $\frac{S_{20} - 10S_{19}}{S_{18}}$ is equal to

52. A curve y = f(x) passing through origin satisfies the differential equation $\frac{9dy}{dx} + \frac{36x^3}{1+x^4}y = \frac{50x^4}{1+x^4}$. If the area enclosed by $y = f^{-1}(x)$, the x-axis and the ordinate $x = \frac{5}{9}$ in Ist quadrant is A, then the value of $\frac{36A}{\pi}$ is

53. Let $f(x) = \begin{cases} \frac{\sin\alpha}{4} + (x-2)^2 & ; \ |x-2| \le k \\ \frac{\cos\alpha}{4} + |x-2| & ; \ |x-2| > k \end{cases}$; k > 0. If f(x) is differentiable at x = 2 – k, 2 + k, then the number of values of $\alpha \in [0, 5\pi]$ is/are

54. Let $\left(1 + x + x^2\right)^{20} = \sum_{k=0}^{40} a_k x^k$. If $\alpha a_{11} = \beta a_{10} + \gamma a_9$, where $\alpha, \beta, \gamma \in N$, then the value of $\frac{\gamma - \alpha}{\beta}$ is